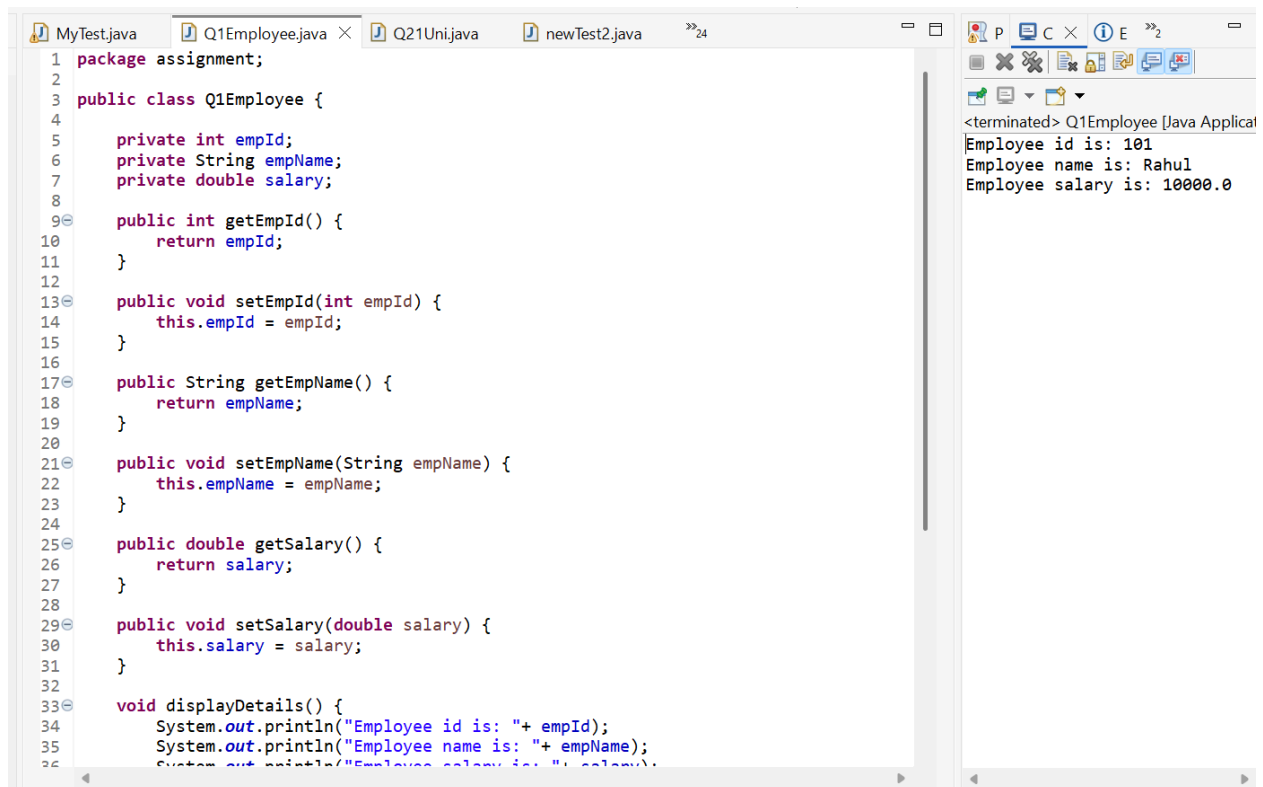
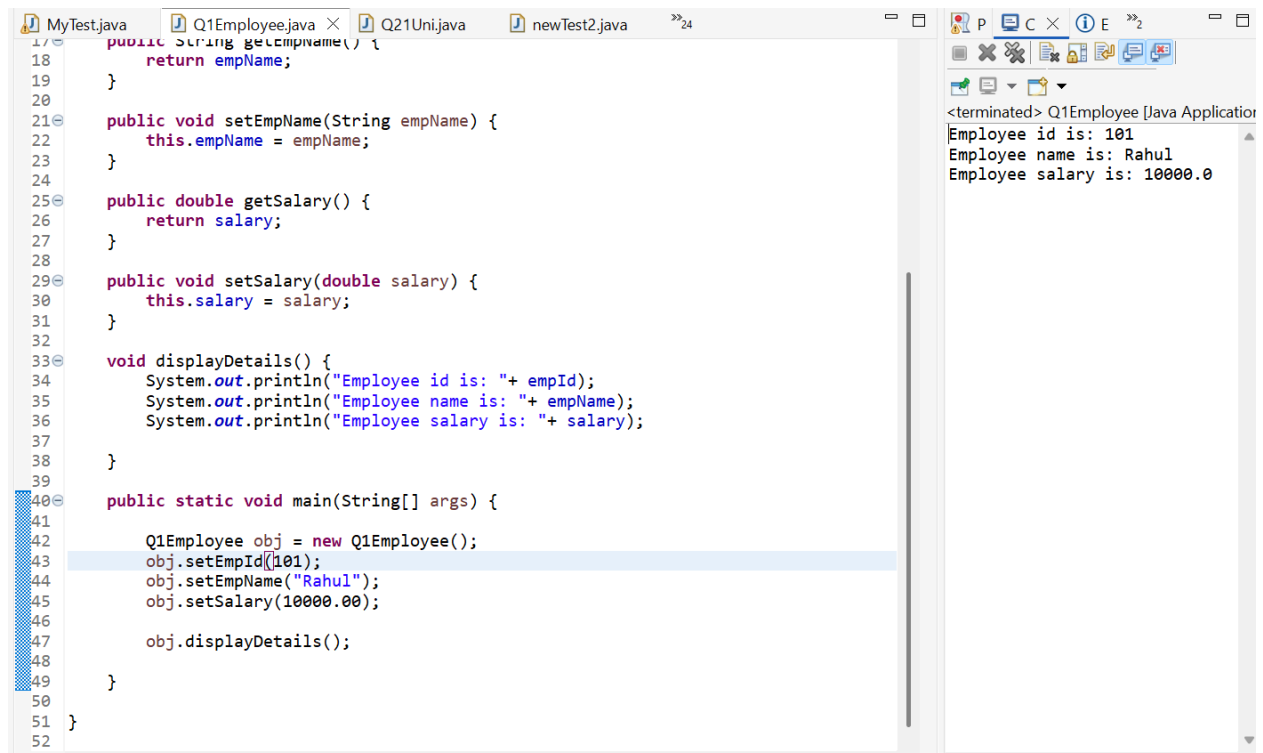


Q1.



```
1 package assignment;
2
3 public class Q1Employee {
4
5     private int empId;
6     private String empName;
7     private double salary;
8
9     public int getEmpId() {
10         return empId;
11     }
12
13     public void setEmpId(int empId) {
14         this.empId = empId;
15     }
16
17     public String getEmpName() {
18         return empName;
19     }
20
21     public void setEmpName(String empName) {
22         this.empName = empName;
23     }
24
25     public double getSalary() {
26         return salary;
27     }
28
29     public void setSalary(double salary) {
30         this.salary = salary;
31     }
32
33     void displayDetails() {
34         System.out.println("Employee id is: " + empId);
35         System.out.println("Employee name is: " + empName);
36         System.out.println("Employee salary is: " + salary);
37     }
38 }
```

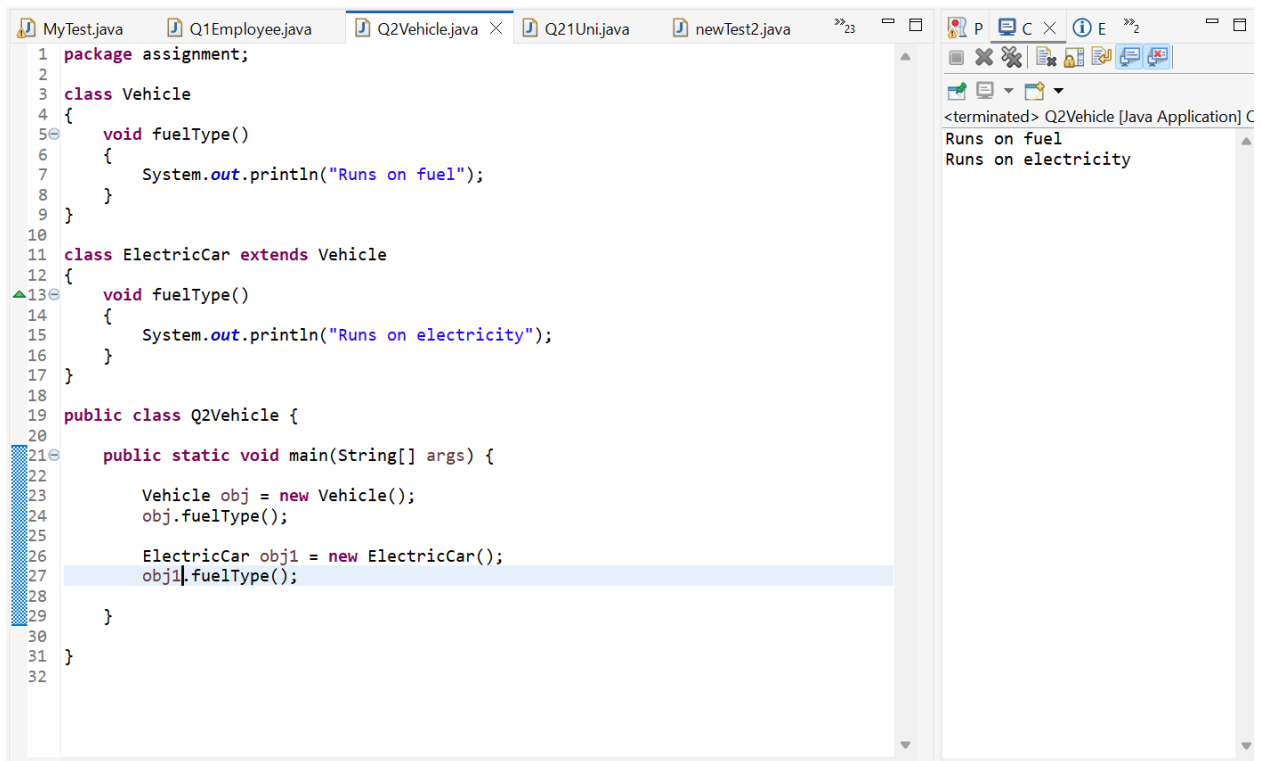
<terminated> Q1Employee [Java Application]
Employee id is: 101
Employee name is: Rahul
Employee salary is: 10000.0



```
17 public String getEmpName() {
18     return empName;
19 }
20
21 public void setEmpName(String empName) {
22     this.empName = empName;
23 }
24
25 public double getSalary() {
26     return salary;
27 }
28
29 public void setSalary(double salary) {
30     this.salary = salary;
31 }
32
33 void displayDetails() {
34     System.out.println("Employee id is: " + empId);
35     System.out.println("Employee name is: " + empName);
36     System.out.println("Employee salary is: " + salary);
37 }
38
39
40 public static void main(String[] args) {
41
42     Q1Employee obj = new Q1Employee();
43     obj.setEmpId(101);
44     obj.setEmpName("Rahul");
45     obj.setSalary(10000.00);
46
47     obj.displayDetails();
48 }
49
50 }
51
52 }
```

<terminated> Q1Employee [Java Application]
Employee id is: 101
Employee name is: Rahul
Employee salary is: 10000.0

Q2.

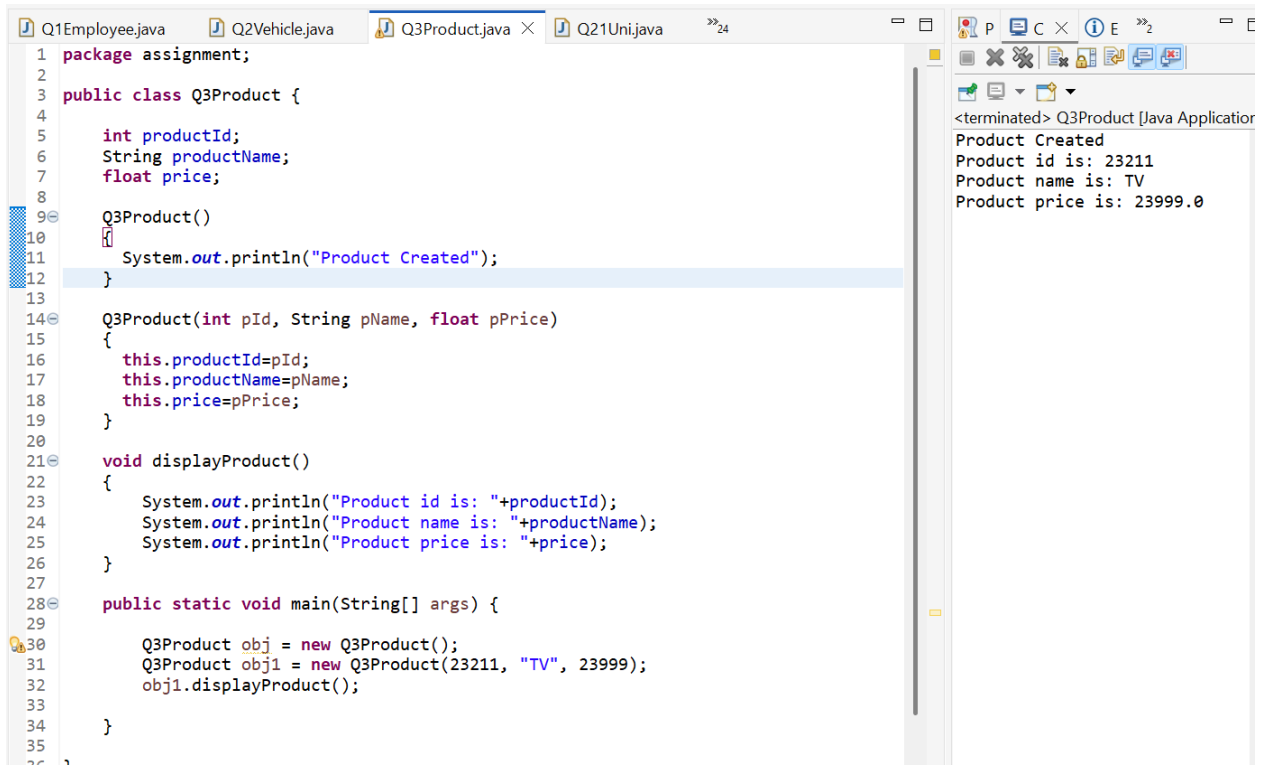


The screenshot shows an IDE with several tabs: MyTest.java, Q1Employee.java, Q2Vehicle.java (active), Q21Uni.java, and newTest2.java. The code in Q2Vehicle.java defines a `Vehicle` class with a `fuelType()` method that prints "Runs on fuel". It also defines an `ElectricCar` class that extends `Vehicle` and overrides `fuelType()` to print "Runs on electricity". A `Q2Vehicle` class contains a `main` method that creates a `Vehicle` object and an `ElectricCar` object, both calling `fuelType()`. The output window on the right shows the results of running the application: "Runs on fuel" followed by "Runs on electricity".

```
1 package assignment;
2
3 class Vehicle
4 {
5     void fuelType()
6     {
7         System.out.println("Runs on fuel");
8     }
9 }
10
11 class ElectricCar extends Vehicle
12 {
13     void fuelType()
14     {
15         System.out.println("Runs on electricity");
16     }
17 }
18
19 public class Q2Vehicle {
20
21     public static void main(String[] args) {
22
23         Vehicle obj = new Vehicle();
24         obj.fuelType();
25
26         ElectricCar obj1 = new ElectricCar();
27         obj1.fuelType();
28
29     }
30 }
31
32
```

<terminated> Q2Vehicle [Java Application] C
Runs on fuel
Runs on electricity

Q3.

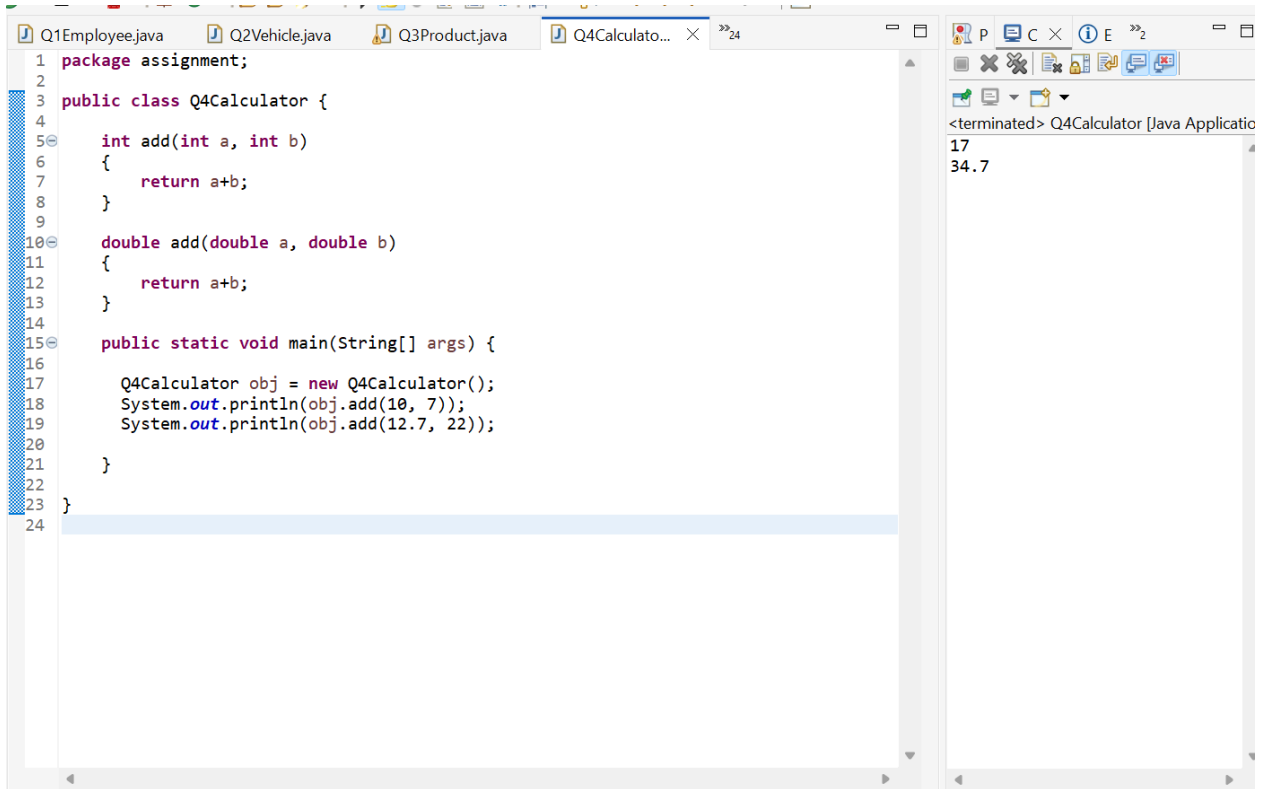


The screenshot shows an IDE with tabs: Q1Employee.java, Q2Vehicle.java, Q3Product.java (active), and Q21Uni.java. The code in Q3Product.java defines a `Q3Product` class with attributes `productId`, `productName`, and `price`. It has a no-argument constructor that prints "Product Created", a parameterized constructor, and a `displayProduct()` method that prints the product details. The `main` method creates two `Q3Product` objects: one with default values and one with specific values (23211, "TV", 23999). The output window on the right shows the results: "Product Created", "Product id is: 23211", "Product name is: TV", and "Product price is: 23999.0".

```
1 package assignment;
2
3 public class Q3Product {
4
5     int productId;
6     String productName;
7     float price;
8
9     Q3Product()
10    {
11        System.out.println("Product Created");
12    }
13
14    Q3Product(int pId, String pName, float pPrice)
15    {
16        this.productId=pId;
17        this.productName=pName;
18        this.price=pPrice;
19    }
20
21    void displayProduct()
22    {
23        System.out.println("Product id is: "+productId);
24        System.out.println("Product name is: "+productName);
25        System.out.println("Product price is: "+price);
26    }
27
28    public static void main(String[] args) {
29
30        Q3Product obj = new Q3Product();
31        Q3Product obj1 = new Q3Product(23211, "TV", 23999);
32        obj1.displayProduct();
33
34    }
35
36
```

<terminated> Q3Product [Java Application]
Product Created
Product id is: 23211
Product name is: TV
Product price is: 23999.0

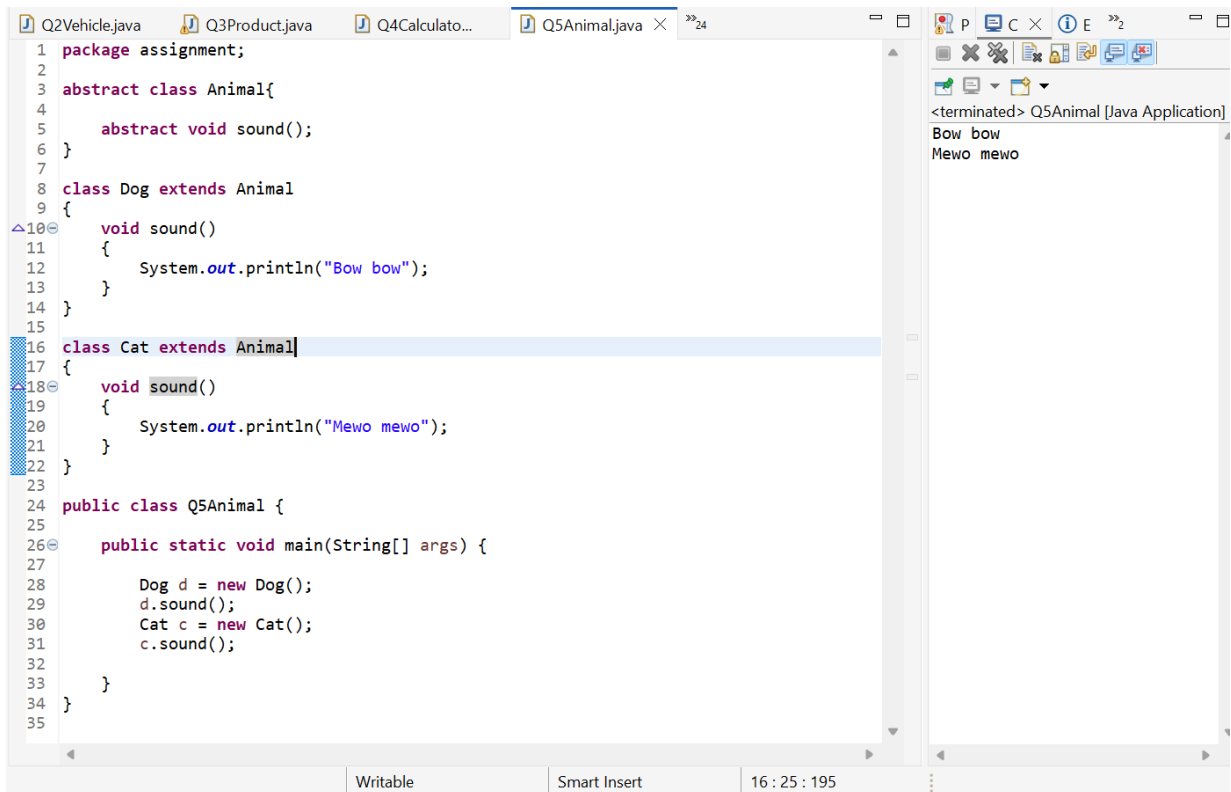
Q4.



```
1 package assignment;
2
3 public class Q4Calculator {
4
5     int add(int a, int b)
6     {
7         return a+b;
8     }
9
10    double add(double a, double b)
11    {
12        return a+b;
13    }
14
15    public static void main(String[] args) {
16
17        Q4Calculator obj = new Q4Calculator();
18        System.out.println(obj.add(10, 7));
19        System.out.println(obj.add(12.7, 22));
20
21    }
22 }
23
24
```

<terminated> Q4Calculator [Java Application]
17
34.7

Q5.

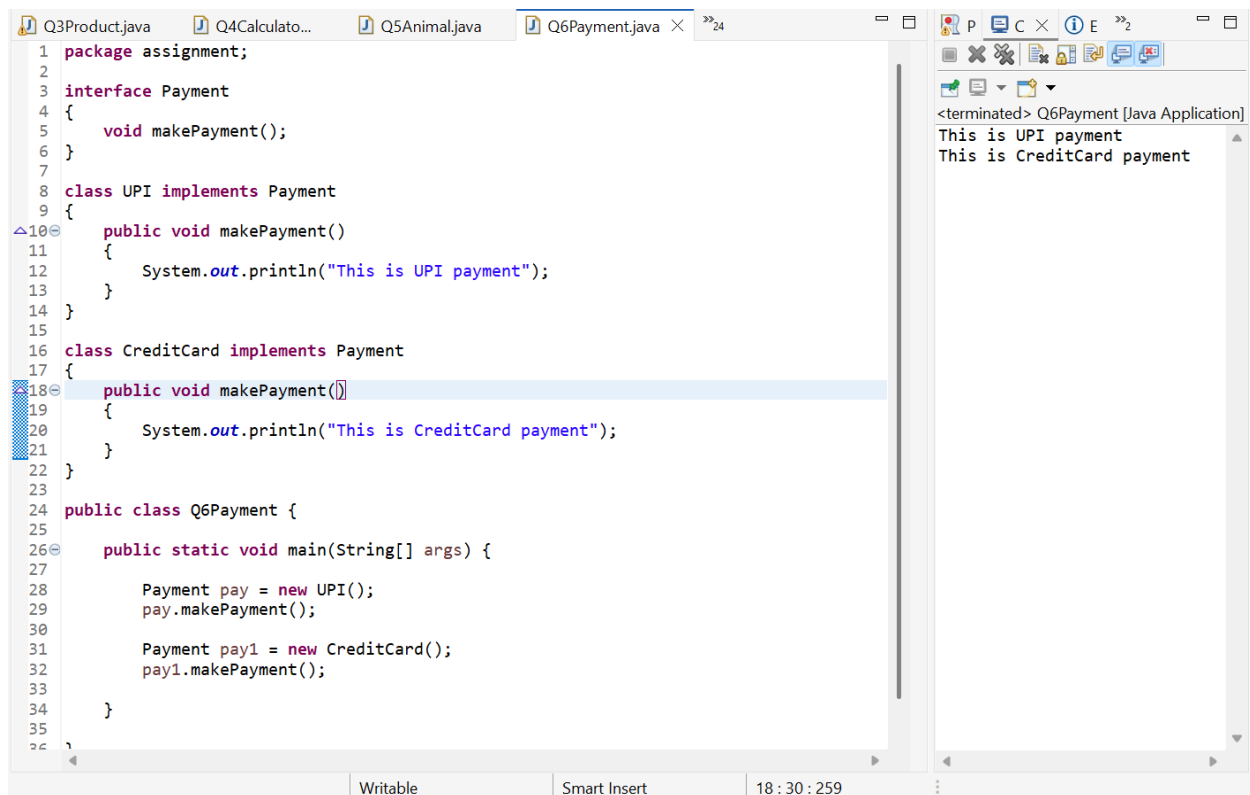


```
1 package assignment;
2
3 abstract class Animal{
4
5     abstract void sound();
6 }
7
8 class Dog extends Animal
9 {
10    void sound()
11    {
12        System.out.println("Bow bow");
13    }
14 }
15
16 class Cat extends Animal{
17 {
18    void sound()
19    {
20        System.out.println("Mewo mewo");
21    }
22 }
23
24 public class Q5Animal {
25
26    public static void main(String[] args) {
27
28        Dog d = new Dog();
29        d.sound();
30        Cat c = new Cat();
31        c.sound();
32
33    }
34 }
35
```

<terminated> Q5Animal [Java Application]
Bow bow
Mewo mewo

Writable Smart Insert 16 : 25 : 195

Q6.

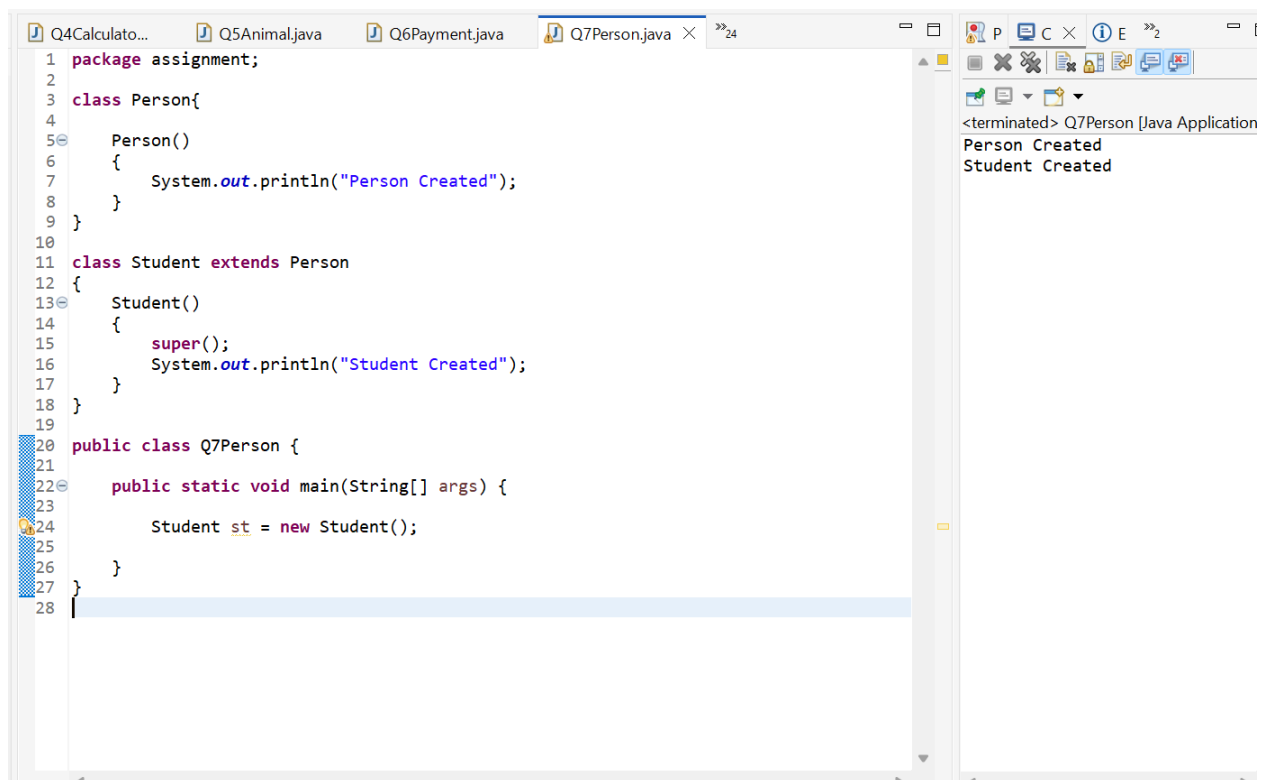


```
1 package assignment;
2
3 interface Payment
4 {
5     void makePayment();
6 }
7
8 class UPI implements Payment
9 {
10     public void makePayment()
11     {
12         System.out.println("This is UPI payment");
13     }
14 }
15
16 class CreditCard implements Payment
17 {
18     public void makePayment()
19     {
20         System.out.println("This is CreditCard payment");
21     }
22 }
23
24 public class Q6Payment {
25
26     public static void main(String[] args) {
27
28         Payment pay = new UPI();
29         pay.makePayment();
30
31         Payment pay1 = new CreditCard();
32         pay1.makePayment();
33     }
34 }
35
36
```

Writable Smart Insert 18:30:259

<terminated> Q6Payment [Java Application]
This is UPI payment
This is CreditCard payment

Q7.

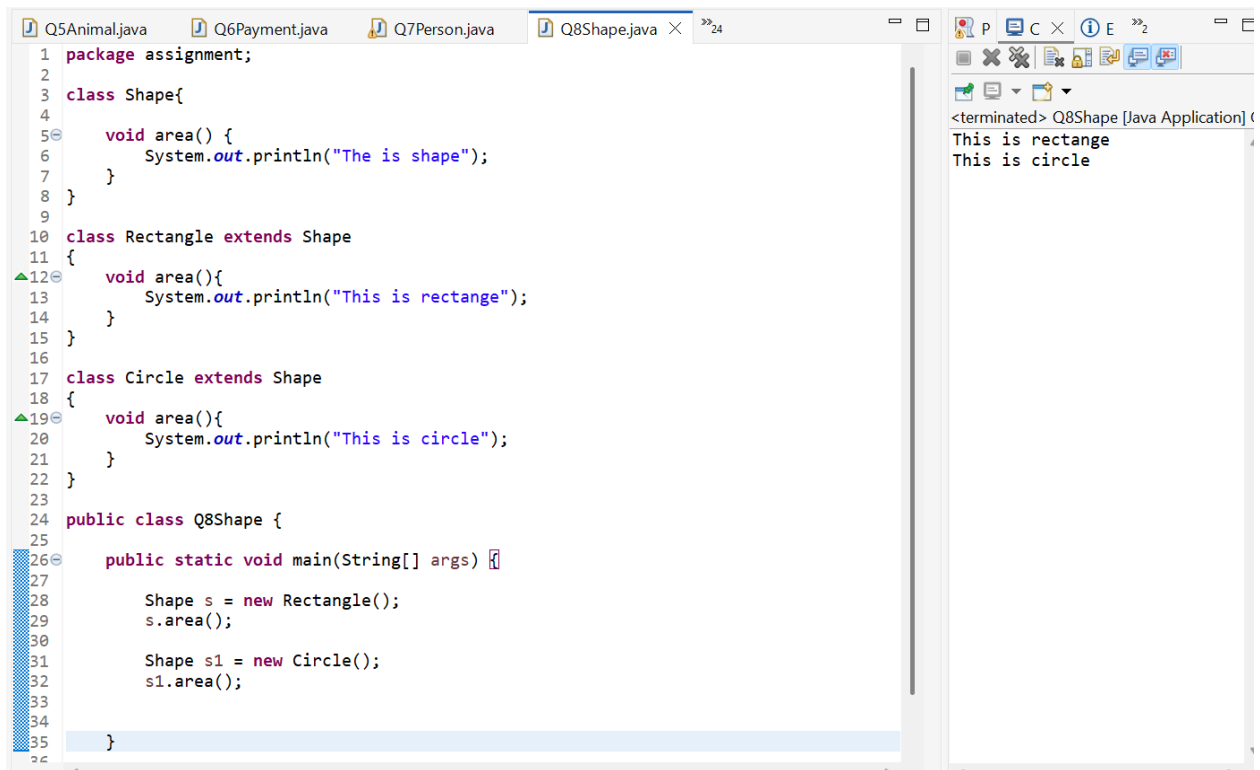


```
1 package assignment;
2
3 class Person{
4
5     Person()
6     {
7         System.out.println("Person Created");
8     }
9 }
10
11 class Student extends Person
12 {
13     Student()
14     {
15         super();
16         System.out.println("Student Created");
17     }
18 }
19
20 public class Q7Person {
21
22     public static void main(String[] args) {
23
24         Student st = new Student();
25
26     }
27 }
28
```

Q4Calculato... Q5Animal.java Q6Payment.java Q7Person.java

<terminated> Q7Person [Java Application]
Person Created
Student Created

Q8.

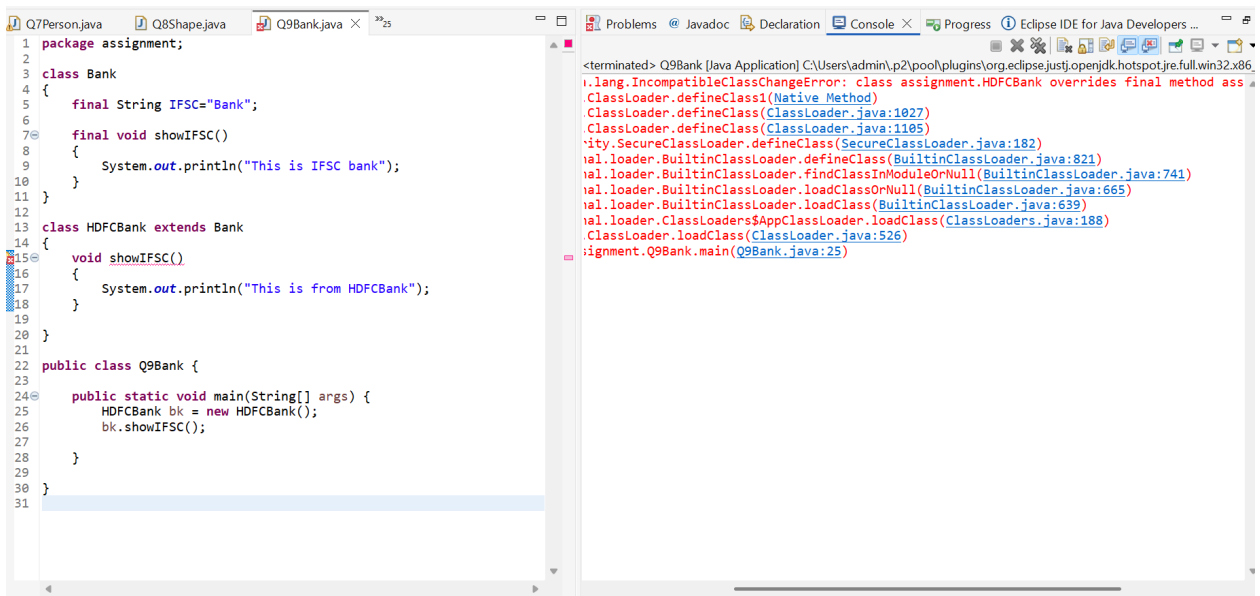


The screenshot shows the Eclipse IDE with the file `Q8Shape.java` open. The code defines a `Shape` class with an `area()` method, and two subclasses, `Rectangle` and `Circle`, each overriding the `area()` method. A `Q8Shape` class contains a `main` method that creates instances of `Rectangle` and `Circle` and calls their `area()` methods. The console on the right shows the output of the program: "This is rectangle" and "This is circle".

```
1 package assignment;
2
3 class Shape{
4
5     void area() {
6         System.out.println("The is shape");
7     }
8 }
9
10 class Rectangle extends Shape
11 {
12     void area(){
13         System.out.println("This is rectange");
14     }
15 }
16
17 class Circle extends Shape
18 {
19     void area(){
20         System.out.println("This is circle");
21     }
22 }
23
24 public class Q8Shape {
25
26     public static void main(String[] args) {
27
28         Shape s = new Rectangle();
29         s.area();
30
31         Shape s1 = new Circle();
32         s1.area();
33
34     }
35 }
```

<terminated> Q8Shape [Java Application] C
This is rectangle
This is circle

Q9.



The screenshot shows the Eclipse IDE with the file `Q9Bank.java` open. The code defines a `Bank` class with a `showIFSC()` method, and a subclass `HDFCBank` that overrides it. A `Q9Bank` class contains a `main` method that creates an instance of `HDFCBank` and calls its `showIFSC()` method. The console on the right shows a `ClassNotFoundException` error, indicating that the `HDFCBank` class is not found.

```
1 package assignment;
2
3 class Bank
4 {
5     final String IFSC="Bank";
6
7     final void showIFSC()
8     {
9         System.out.println("This is IFSC bank");
10    }
11 }
12
13 class HDFCBank extends Bank
14 {
15     void showIFSC()
16     {
17         System.out.println("This is from HDFCBank");
18     }
19 }
20
21
22 public class Q9Bank {
23
24     public static void main(String[] args) {
25         HDFCBank bk = new HDFCBank();
26         bk.showIFSC();
27     }
28 }
29
30
31 }
```

<terminated> Q9Bank [Java Application] C:\Users\admin\p2\pool\plugins\org.eclipse.justi.openjdk.hotspot.jre.full.win32.x86_64.jre\bin\java.exe
java.lang.ClassNotFoundException: class assignment.HDFCBank overrides final method ass
ClassLoader.defineClass(Native Method)
ClassLoader.defineClass(ClassLoader.java:1027)
ClassLoader.defineClass(ClassLoader.java:1105)
ity.SecureClassLoader.defineClass(SecureClassLoader.java:182)
al.loader.BuiltinClassLoader.defineClass(BuiltinClassLoader.java:821)
al.loader.BuiltinClassLoader.findClassInModuleOrNull(BuiltinClassLoader.java:741)
al.loader.BuiltinClassLoader.loadClassOrNull(BuiltinClassLoader.java:665)
al.loader.BuiltinClassLoader.loadClass(BuiltinClassLoader.java:639)
al.loader.ClassLoaders\$AppClassLoader.loadClass(ClassLoaders.java:188)
ClassLoader.loadClass(ClassLoader.java:526)
ignment.Q9Bank.main(Q9Bank.java:25)

Q10.

```

1 package assignment;
2
3 class CStudent
4 {
5     static String collegeName="Dayananda sagar";
6     String name;
7     int rollNo;
8
9     CStudent(String name, int rollNo) {
10         this.name = name;
11         this.rollNo = rollNo;
12     }
13
14     void display()
15     {
16         System.out.println("College name is: "+collegeName);
17         System.out.println("Student name is: "+name);
18         System.out.println("Student roll number is: "+rollNo);
19         System.out.println("-----");
20     }
21 }
22
23 public class Q10Student {
24
25     public static void main(String[] args) {
26
27         CStudent s1 = new CStudent("Alice", 101);
28         CStudent s2 = new CStudent("Bob", 102);
29         CStudent s3 = new CStudent("Charlie", 103);
30
31         s1.display();
32         s2.display();
33         s3.display();
34
35         CStudent collegeName = "XYZ College";
36     }
37 }

```

```

<terminated> Q10Student [Java Application] C:\Users\admin\p2\pool\pl
College name is: Dayananda sagar
Student name is: Alice
Student roll number is: 101
-----
College name is: Dayananda sagar
Student name is: Bob
Student roll number is: 102
-----
College name is: Dayananda sagar
Student name is: Charlie
Student roll number is: 103
-----
After changing college name:
College name is: XYZ College
Student name is: Alice
Student roll number is: 101
-----
College name is: XYZ College
Student name is: Bob
Student roll number is: 102
-----
College name is: XYZ College
Student name is: Charlie
Student roll number is: 103
-----

```

Q11.

```

1 package assignment;
2
3 class Library
4 {
5     String libraryName;
6
7     Library()
8     {
9         System.out.println("Welcome to the Library!");
10    }
11
12    void showLocation()
13    {
14        System.out.println("This library is located in Mumbai");
15    }
16 }
17
18 public class Q11Library {
19
20     public static void main(String[] args) {
21
22         Library l = new Library();
23         l.showLocation();
24     }
25 }
26
27

```

```

<terminated> Q11Library [Java Application] C:\Users\admin\p2\pool\p
Welcome to the Library!
This library is located in Mumbai

```

Q12.

```
1 package assignment;
2
3 class Account{
4     private String accountHolderName;
5     private float balance;
6
7     public String getAccountHolderName() {
8         return accountHolderName;
9     }
10
11     public void setAccountHolderName(String accountHolderName) {
12         this.accountHolderName = accountHolderName;
13     }
14
15     public float getBalance() {
16         return balance;
17     }
18
19     public void setBalance(float balance) {
20
21         if (balance<0)
22         {
23             System.out.println("Please do not enter negative number");
24         }
25         else
26         {
27             this.balance = balance;
28         }
29     }
30 }
31
32 }
33
34 public class Q12Account {
35
36     public static void main(String[] args) {
```

<terminated> Q12Account [Java Application] C:\Users\admin\p2\pool\p
Bharat BJ
9.0

Writable Smart Insert 42 : 45 : 816

```
Q9Bank.java Q10Student.java Q11Library.java Q12Account.java x 24
13 this.accountHolderName = accountHolderName;
14 }
15
16 public float getBalance() {
17     return balance;
18 }
19
20 public void setBalance(float balance) {
21
22     if (balance<0)
23     {
24         System.out.println("Please do not enter negative number");
25     }
26     else
27     {
28         this.balance = balance;
29     }
30 }
31
32 }
33
34 public class Q12Account {
35
36     public static void main(String[] args) {
37
38         Account ac = new Account();
39         ac.setAccountHolderName("Bharat BJ");
40         ac.setBalance(9);
41         System.out.println(ac.getAccountHolderName());
42         System.out.println(ac.getBalance());
43     }
44 }
45
46 }
47
```

<terminated> Q12Account [Java Application] C:\Users\admin\p2\pool\p
Bharat BJ
9.0

Writable Smart Insert 42 : 45 : 816

Q13.

The screenshot shows an IDE with a Java file named Q13Inherit.java. The code defines a hierarchy of classes: Device, Mobile (extending Device), and SmartPhone (extending Mobile). A main method in Q13Inherit creates a SmartPhone object and calls its start(), calling(), and internet() methods. The output window shows the execution results: SmartPhone, Mobile, and Device, each on a new line.

```
1 package assignment;
2
3 class Device{
4     void start()
5     {
6         System.out.println("Device");
7     }
8 }
9
10 class Mobile extends Device {
11     void calling()
12     {
13         System.out.println("Mobile");
14     }
15 }
16
17 class SmartPhone extends Mobile{
18     void internet()
19     {
20         System.out.println("SmartPhone");
21     }
22 }
23
24 public class Q13Inherit {
25
26     public static void main(String[] args) {
27
28         SmartPhone sm = new SmartPhone();
29         sm.internet();
30         sm.calling();
31         sm.start();
32     }
33 }
34
35
```

Output:

```
<terminated> Q13Inherit [Java Application] C:\Users\admin\p2\pool\plu
SmartPhone
Mobile
Device
```

Q14.

The screenshot shows an IDE with a Java file named Q14HInherit.java. The code defines a hierarchy of classes: Course, Science (extending Course), Commerce (extending Course), and Arts (extending Course). A main method in Q14HInherit creates a Course object and calls its courseInfo() method. The output window shows the execution results: Your course____Science, Your course____Commerce, and Your course____Arts, each on a new line.

```
1 package assignment;
2
3 class Course{
4     void courseInfo()
5     {
6         System.out.println("____Your course____");
7     }
8 }
9
10 class Science extends Course{
11     void courseInfo()
12     {
13         System.out.println("Your course____Science");
14     }
15 }
16
17 class Commerce extends Course
18 {
19     void courseInfo()
20     {
21         System.out.println("Your course____Commerce");
22     }
23 }
24 class Arts extends Course
25 {
26     void courseInfo()
27     {
28         System.out.println("Your course____Arts");
29     }
30 }
31
32 public class Q14HInherit {
33
34     public static void main(String[] args) {
35
36         Course c = new Course();
37     }
38 }
39
```

Output:

```
<terminated> Q14HInherit [Java Application] C:\Users\admin\p2\pool\pl
____Your course____
Your course____Science
Your course____Commerce
Your course____Arts
```



```
Q11Library.java Q12Account.java Q13Inherit.java Q14HInherit... × 24
13 }
14 }
15 }
16 }
17 class Commerce extends Course
18 {
19     void courseInfo()
20     {
21         System.out.println("Your course____Commerce");
22     }
23 }
24 class Arts extends Course
25 {
26     void courseInfo()
27     {
28         System.out.println("Your course____Arts");
29     }
30 }
31
32 public class Q14HInherit {
33
34     public static void main(String[] args) {
35
36         Course c = new Course();
37         c.courseInfo();
38         Science s = new Science();
39         s.courseInfo();
40         Commerce com = new Commerce();
41         com.courseInfo();
42         Arts a = new Arts();
43         a.courseInfo();
44     }
45 }
46
47 }
48 }
```

<terminated> Q14HInherit [Java Application] C:\Users\admin\p2\pool\p1
Your course____
Your course____Science
Your course____Commerce
Your course____Arts

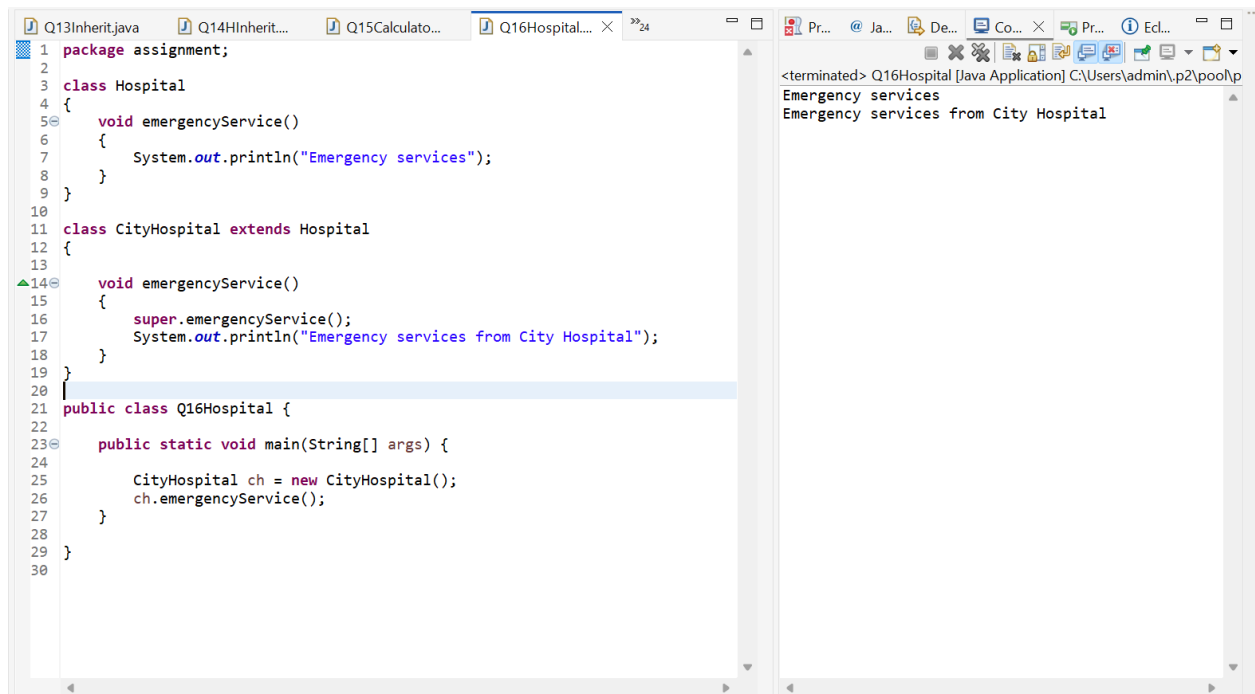
Q15.

```
Q12Account.java Q13Inherit.java Q14HInherit... Q15Calculato... × 24
1 package assignment;
2
3 class LoanCalculator
4 {
5     int amount;
6     double interestRate;
7
8     void calculateLoan(int amount)
9     {
10         this.amount = amount;
11         System.out.println("Loan amount: "+amount);
12         System.out.println("Loan interest rate: "+interestRate);
13         System.out.println("_____");
14     }
15
16     void calculateLoan(int amount, double interestRate)
17     {
18         this.amount = amount;
19         this.interestRate = interestRate;
20         System.out.println("Loan amount: "+amount);
21         System.out.println("Loan interest rate: "+interestRate);
22     }
23 }
24
25
26
27 public class Q15Calculator {
28
29     public static void main(String[] args) {
30
31         LoanCalculator lc = new LoanCalculator();
32         lc.calculateLoan(29000);
33         lc.calculateLoan(17000, 7.99);
34     }
35
36 }
```

<terminated> Q15Calculator [Java Application] C:\Users\admin\p2\pool\h
Loan amount: 29000
Loan interest rate: 0.0

Loan amount: 17000
Loan interest rate: 7.99

Q16.

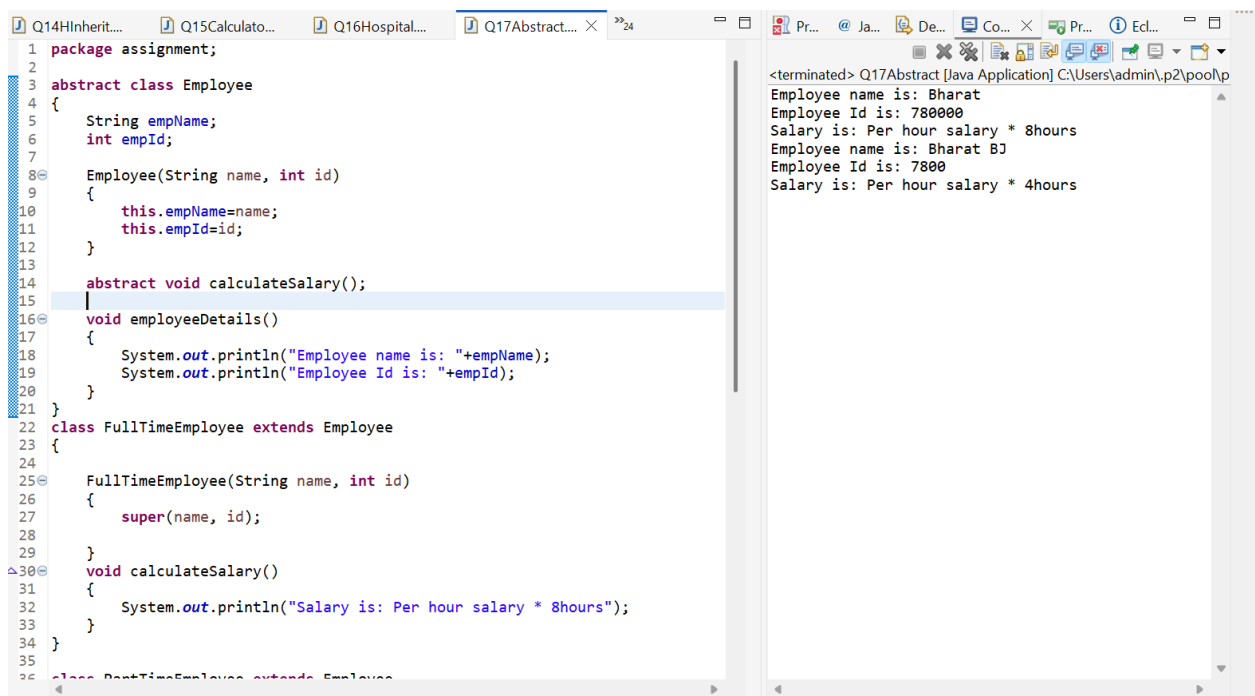


The screenshot shows an IDE with a Java file named Q16Hospital.java. The code defines a base class Hospital with an emergencyService() method that prints "Emergency services". A subclass CityHospital extends Hospital and overrides emergencyService() to print "Emergency services from City Hospital". A public class Q16Hospital contains a main method that creates a CityHospital object and calls its emergencyService() method. The output window on the right shows the result of the program execution.

```
1 package assignment;
2
3 class Hospital
4 {
5     void emergencyService()
6     {
7         System.out.println("Emergency services");
8     }
9 }
10
11 class CityHospital extends Hospital
12 {
13
14     void emergencyService()
15     {
16         super.emergencyService();
17         System.out.println("Emergency services from City Hospital");
18     }
19 }
20
21 public class Q16Hospital {
22
23     public static void main(String[] args) {
24
25         CityHospital ch = new CityHospital();
26         ch.emergencyService();
27     }
28 }
29
30
```

<terminated> Q16Hospital [Java Application] C:\Users\admin\p2\pool\p
Emergency services
Emergency services from City Hospital

Q17.



The screenshot shows an IDE with a Java file named Q17Abstract.java. The code defines an abstract class Employee with attributes empName and empId, a constructor, and an abstract method calculateSalary(). A subclass FullTimeEmployee extends Employee and implements calculateSalary() to print "Salary is: Per hour salary * 8hours". The output window on the right shows the result of the program execution.

```
1 package assignment;
2
3 abstract class Employee
4 {
5     String empName;
6     int empId;
7
8     Employee(String name, int id)
9     {
10         this.empName=name;
11         this.empId=id;
12     }
13
14     abstract void calculateSalary();
15
16     void employeeDetails()
17     {
18         System.out.println("Employee name is: "+empName);
19         System.out.println("Employee Id is: "+empId);
20     }
21 }
22
23 class FullTimeEmployee extends Employee
24 {
25     FullTimeEmployee(String name, int id)
26     {
27         super(name, id);
28     }
29
30     void calculateSalary()
31     {
32         System.out.println("Salary is: Per hour salary * 8hours");
33     }
34 }
35
36 class PartTimeEmployee extends Employee
37 {
38     PartTimeEmployee(String name, int id)
39     {
40         super(name, id);
41     }
42
43     void calculateSalary()
44     {
45         System.out.println("Salary is: Per hour salary * 4hours");
46     }
47 }
48
```

<terminated> Q17Abstract [Java Application] C:\Users\admin\p2\pool\p
Employee name is: Bharat
Employee Id is: 780000
Salary is: Per hour salary * 8hours
Employee name is: Bharat B1
Employee Id is: 7800
Salary is: Per hour salary * 4hours

```
Q14HInherit... Q15Calculato... Q16Hospital... Q17Abstract... × »24
30  }
31  void calculateSalary()
32  {
33      System.out.println("Salary is: Per hour salary * 8hours");
34  }
35  }
36  class PartTimeEmployee extends Employee
37  {
38      PartTimeEmployee(String name, int id)
39      {
40          super(name, id);
41      }
42  }
43  }
44  void calculateSalary()
45  {
46      System.out.println("Salary is: Per hour salary * 4hours");
47  }
48  }
49  }
50  }
51  public class Q17Abstract {
52  }
53  public static void main(String[] args)
54  {
55      Employee e = new FullTimeEmployee("Bharat", 780000);
56      e.employeeDetails();
57      e.calculateSalary();
58  }
59  Employee e1 = new PartTimeEmployee("Bharat BJ", 7800);
60  e1.employeeDetails();
61  e1.calculateSalary();
62  }
63  }
64  }
```

<terminated> Q17Abstract [Java Application] C:\Users\admin\p2\poolp
Employee name is: Bharat
Employee Id is: 780000
Salary is: Per hour salary * 8hours
Employee name is: Bharat BJ
Employee Id is: 7800
Salary is: Per hour salary * 4hours

Writable Smart Insert 15 : 5 : 204

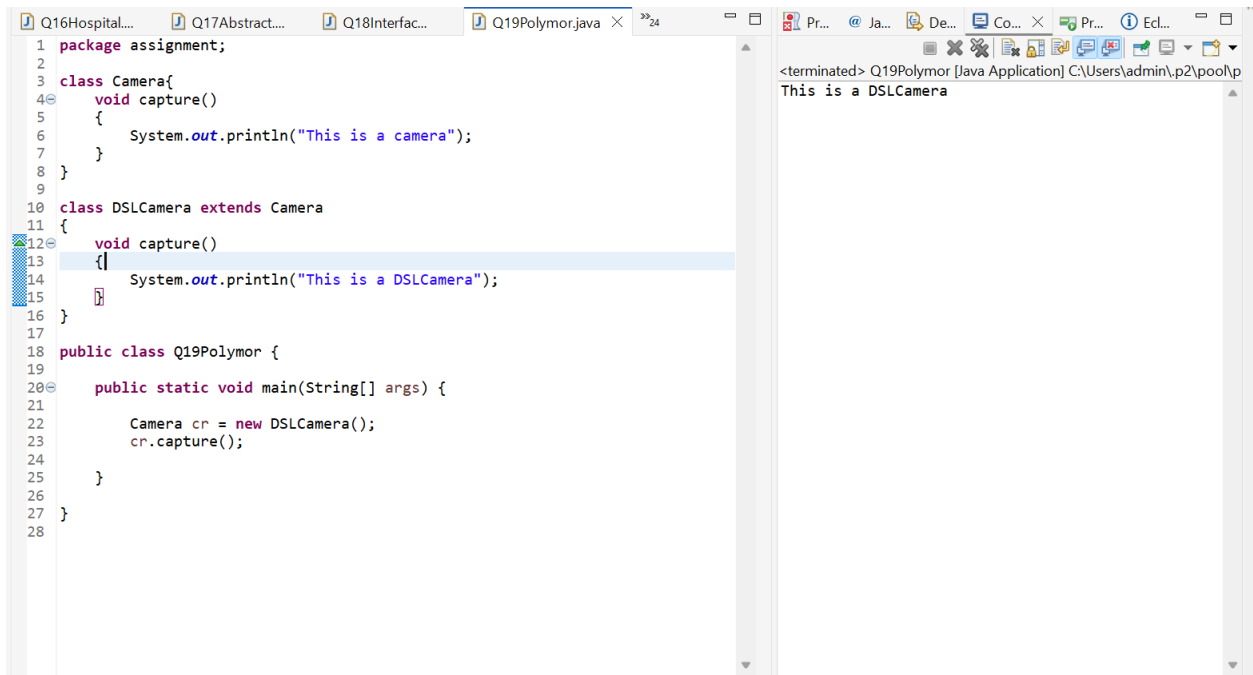
Q18.

```
Q15Calculato... Q16Hospital... Q17Abstract... Q18Interfac... × »24
1  package assignment;
2
3  interface Transport
4  {
5      void booking();
6  }
7
8  class Bus implements Transport
9  {
10     public void booking()
11     {
12         System.out.println("This is bus service");
13     }
14 }
15
16 class Flight implements Transport
17 {
18     public void booking()
19     {
20         System.out.println("This is flight service");
21     }
22 }
23
24
25 public class Q18Interface {
26
27     public static void main(String[] args) {
28
29         Transport tr = new Bus();
30         tr.booking();
31         Transport tr1 = new Flight();
32         tr1.booking();
33     }
34 }
35
36 }
```

<terminated> Q18Interface [Java Application] C:\Users\admin\p2\poolp
This is bus service
This is flight service

Writable Smart Insert 24 : 1 : 313

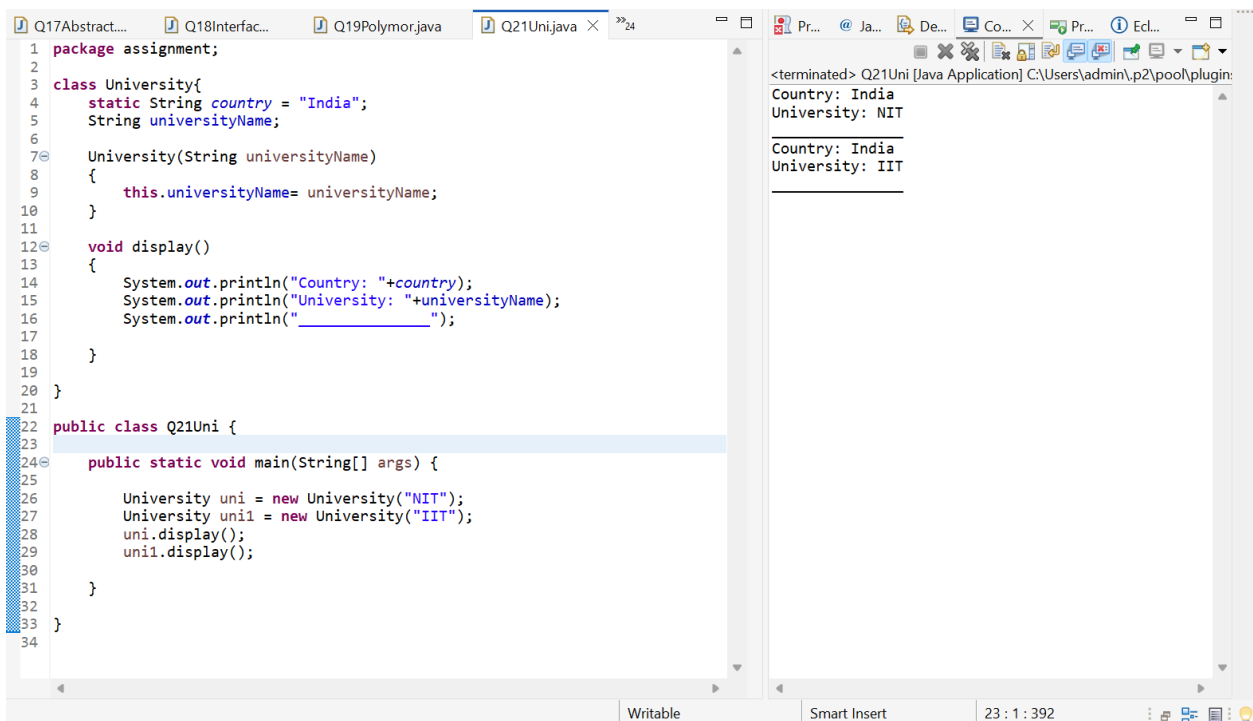
Q19.



```
1 package assignment;
2
3 class Camera{
4     void capture()
5     {
6         System.out.println("This is a camera");
7     }
8 }
9
10 class DSLCamera extends Camera
11 {
12     void capture()
13     {
14         System.out.println("This is a DSLCamera");
15     }
16 }
17
18 public class Q19Polymor {
19
20     public static void main(String[] args) {
21
22         Camera cr = new DSLCamera();
23         cr.capture();
24     }
25 }
26
27 }
28
```

<terminated> Q19Polymor [Java Application] C:\Users\admin\p2\pool\p
This is a DSLCamera

Q21.

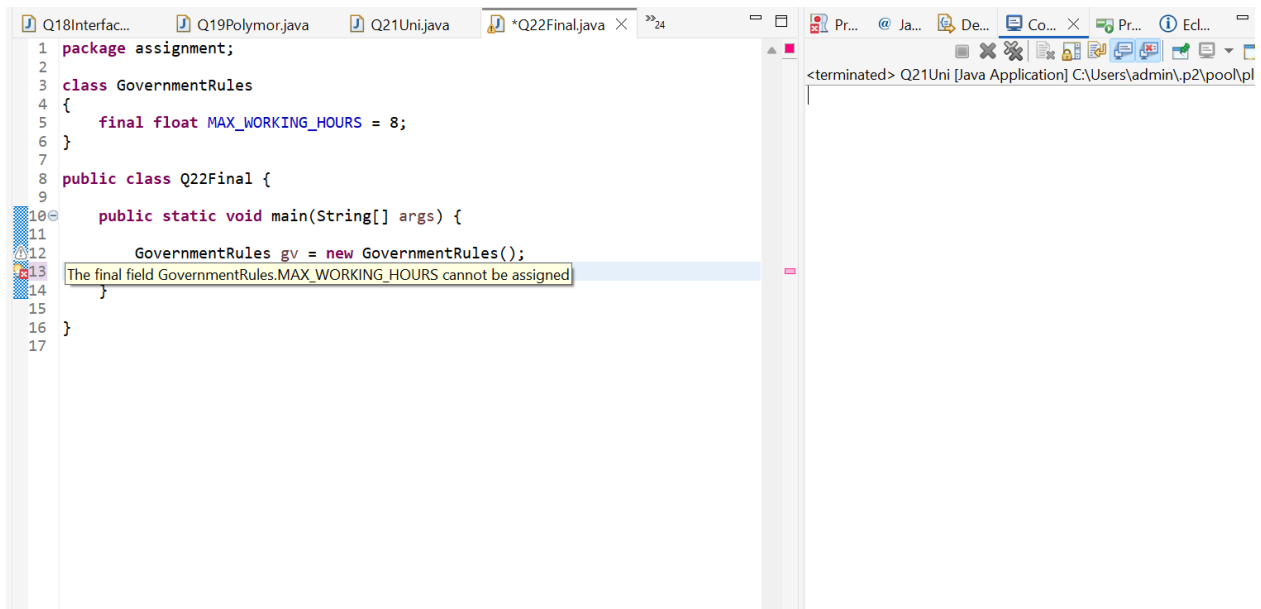


```
1 package assignment;
2
3 class University{
4     static String country = "India";
5     String universityName;
6
7     University(String universityName)
8     {
9         this.universityName= universityName;
10    }
11
12    void display()
13    {
14        System.out.println("Country: "+country);
15        System.out.println("University: "+universityName);
16        System.out.println("_____");
17    }
18 }
19
20 }
21
22 public class Q21Uni {
23
24     public static void main(String[] args) {
25
26         University uni = new University("NIT");
27         University uni1 = new University("IIT");
28         uni.display();
29         uni1.display();
30     }
31 }
32
33 }
34
```

<terminated> Q21Uni [Java Application] C:\Users\admin\p2\pool\plugin:
Country: India
University: NIT

Country: India
University: IIT

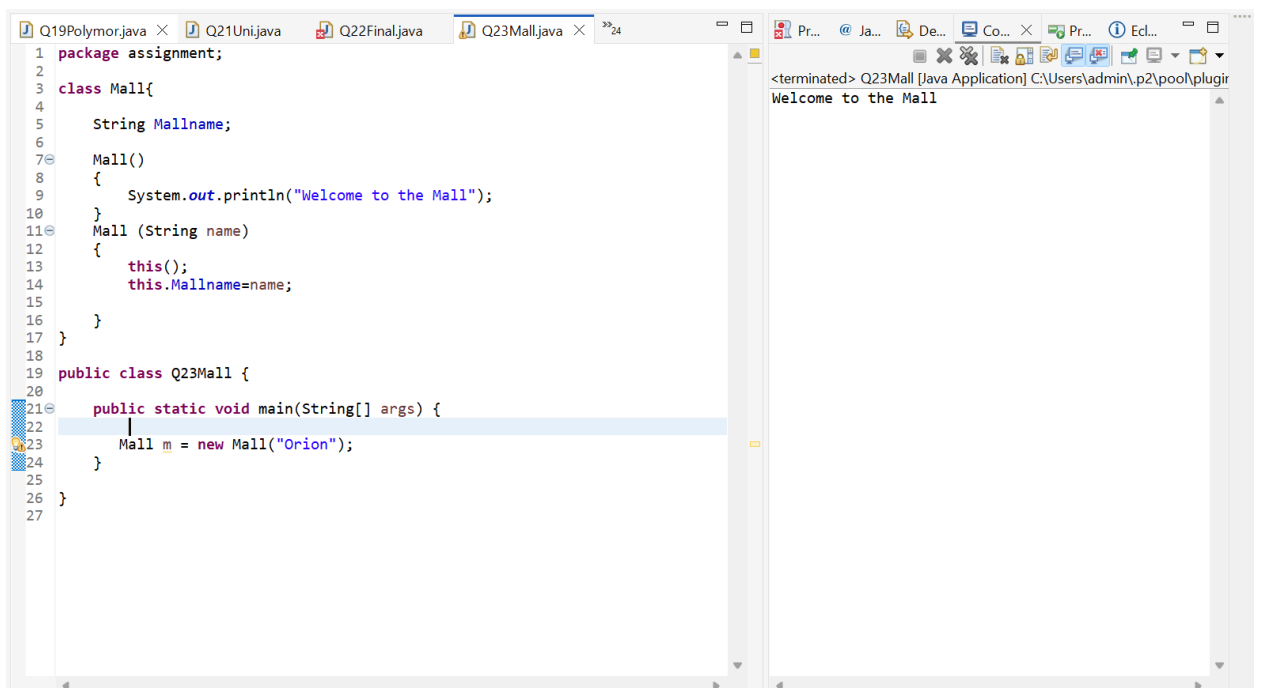
Q22.



```
1 package assignment;
2
3 class GovernmentRules
4 {
5     final float MAX_WORKING_HOURS = 8;
6 }
7
8 public class Q22Final {
9
10     public static void main(String[] args) {
11
12         GovernmentRules gv = new GovernmentRules();
13         GovernmentRules.MAX_WORKING_HOURS = 10;
14     }
15 }
16
17 }
```

<terminated> Q21Uni [Java Application] C:\Users\admin\p2\pool\pl

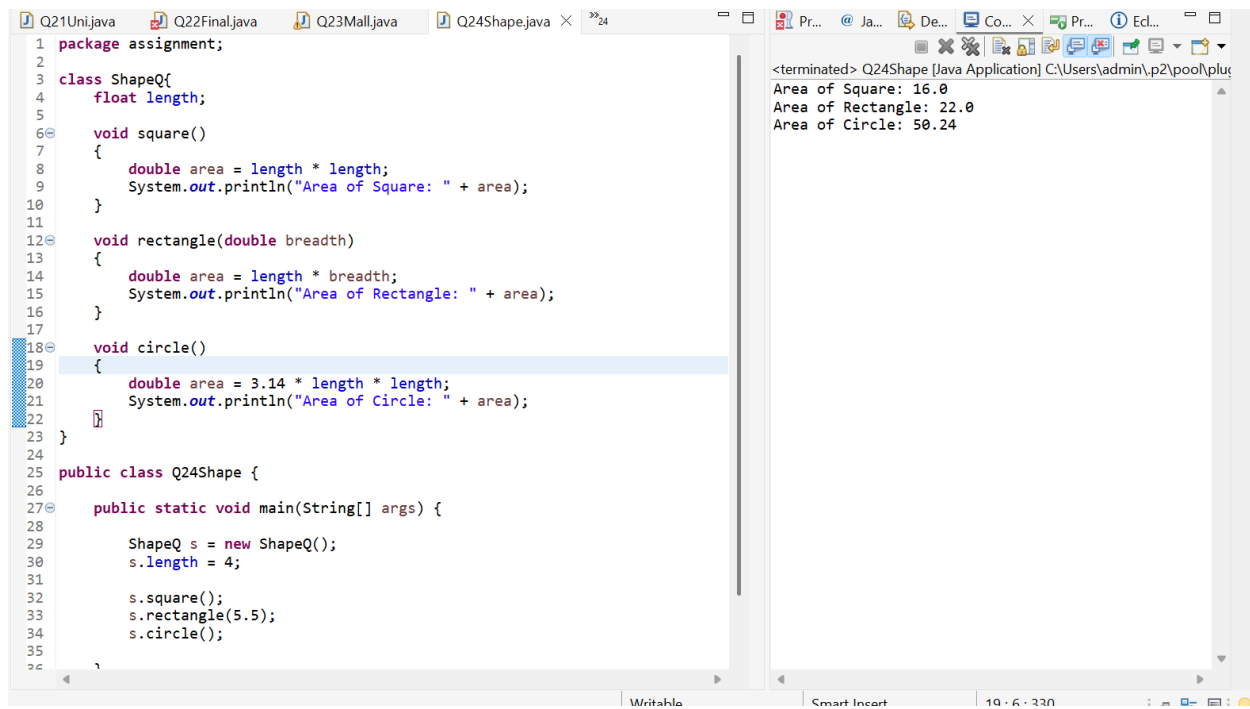
Q23.



```
1 package assignment;
2
3 class Mall{
4
5     String Mallname;
6
7     Mall()
8     {
9         System.out.println("Welcome to the Mall");
10    }
11    Mall (String name)
12    {
13        this();
14        this.Mallname=name;
15    }
16 }
17
18
19 public class Q23Mall {
20
21     public static void main(String[] args) {
22
23         Mall m = new Mall("Orion");
24     }
25 }
26
27 }
```

<terminated> Q23Mall [Java Application] C:\Users\admin\p2\pool\plugir
Welcome to the Mall

Q24.

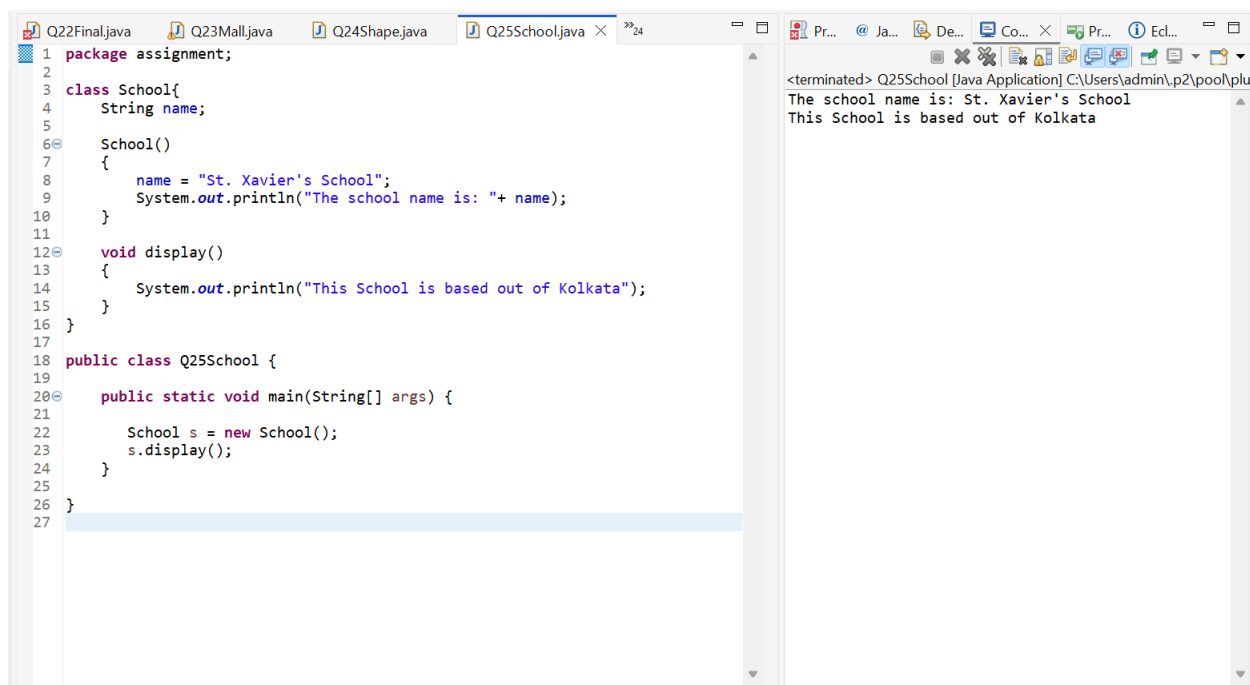


The screenshot shows an IDE with a tab for Q24Shape.java. The code defines a ShapeQ class with methods for calculating the area of a square, rectangle, and circle. A public class Q24Shape contains a main method that creates a ShapeQ object, sets its length to 4, and calls the area calculation methods. The output window on the right shows the results of these calculations.

```
1 package assignment;
2
3 class ShapeQ{
4     float length;
5
6     void square()
7     {
8         double area = length * length;
9         System.out.println("Area of Square: " + area);
10    }
11
12    void rectangle(double breadth)
13    {
14        double area = length * breadth;
15        System.out.println("Area of Rectangle: " + area);
16    }
17
18    void circle()
19    {
20        double area = 3.14 * length * length;
21        System.out.println("Area of Circle: " + area);
22    }
23 }
24
25 public class Q24Shape {
26
27     public static void main(String[] args) {
28
29         ShapeQ s = new ShapeQ();
30         s.length = 4;
31
32         s.square();
33         s.rectangle(5.5);
34         s.circle();
35     }
36 }
```

<terminated> Q24Shape [Java Application] C:\Users\admin\p2\pool\plu
Area of Square: 16.0
Area of Rectangle: 22.0
Area of Circle: 50.24

Q25.

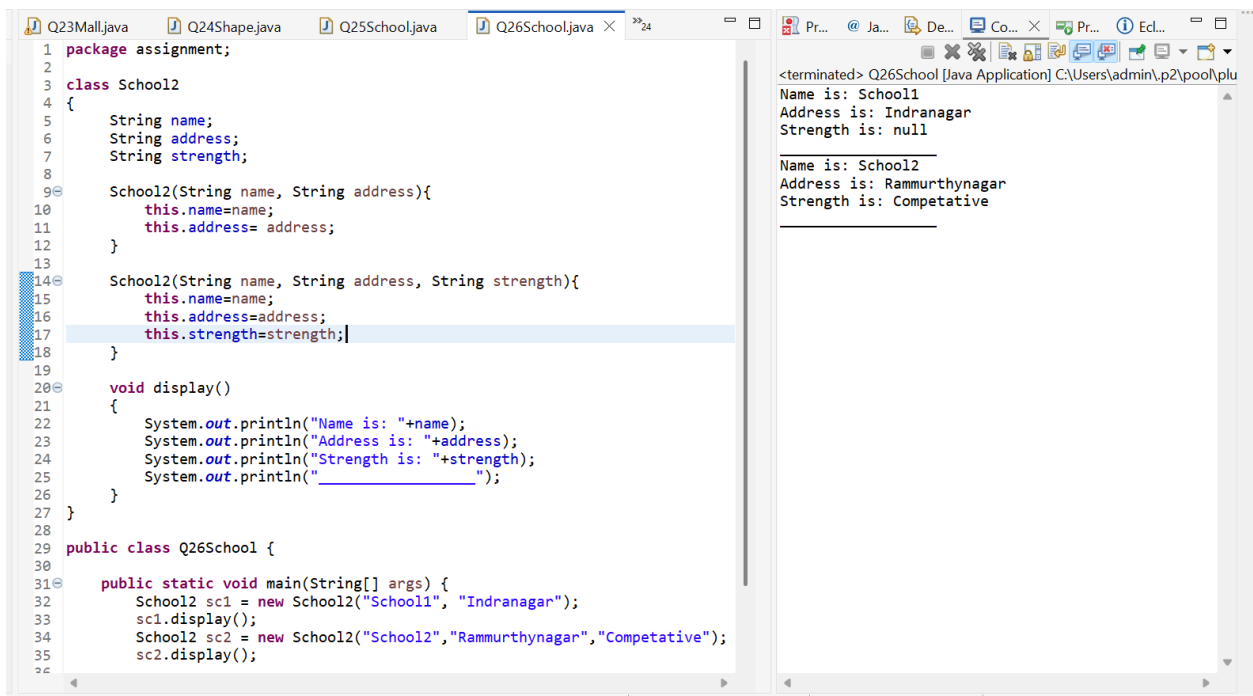


The screenshot shows an IDE with a tab for Q25School.java. The code defines a School class with a name attribute and methods for displaying the school name and location. A public class Q25School contains a main method that creates a School object and calls the display methods. The output window on the right shows the results of these calls.

```
1 package assignment;
2
3 class School{
4     String name;
5
6     School()
7     {
8         name = "St. Xavier's School";
9         System.out.println("The school name is: " + name);
10    }
11
12    void display()
13    {
14        System.out.println("This School is based out of Kolkata");
15    }
16 }
17
18 public class Q25School {
19
20     public static void main(String[] args) {
21
22         School s = new School();
23         s.display();
24     }
25 }
26
27
```

<terminated> Q25School [Java Application] C:\Users\admin\p2\pool\plu
The school name is: St. Xavier's School
This School is based out of Kolkata

Q26.



```
1 package assignment;
2
3 class School2
4 {
5     String name;
6     String address;
7     String strength;
8
9     School2(String name, String address){
10         this.name=name;
11         this.address= address;
12     }
13
14     School2(String name, String address, String strength){
15         this.name=name;
16         this.address=address;
17         this.strength=strength;
18     }
19
20     void display()
21     {
22         System.out.println("Name is: "+name);
23         System.out.println("Address is: "+address);
24         System.out.println("Strength is: "+strength);
25         System.out.println("_____");
26     }
27 }
28
29 public class Q26School {
30
31     public static void main(String[] args) {
32         School2 sc1 = new School2("School1", "Indranagar");
33         sc1.display();
34         School2 sc2 = new School2("School2", "Rammurthynagar", "Competative");
35         sc2.display();
36     }
37 }
```

<terminated> Q26School [Java Application] C:\Users\admin\p2\pool\plu
Name is: School1
Address is: Indranagar
Strength is: null

Name is: School2
Address is: Rammurthynagar
Strength is: Competative
